DocVerify — Final Phase 7 Handover Report

Final System Validation, Production Finalization & Project Submission Readiness

Project: DocVerify AI	**Phase:** Phase 7 (Final Validation & Handover)
Role: Senior ML, CV & Security Research Engineer	**Status:** COMPLETE & PRODUCTION-READY
Handover Date: October 17, 2026	**Phase 6 Regression:** 10/10 Passed | **Phase 7:** 11/11 Passed

1. Executive Status & Final Acceptance

Phase 7 concludes the development and scientific hardening lifecycle of **DocVerify**. The production runtime (app.py and src/inference_engine.py) has been thoroughly audited, upgraded, and verified against the Phase 6 scientific benchmark. All legacy Phase 4 heuristics and disconnected calibration grids were deprecated in favor of native Phase 6 multi-modal fusion.

• **Phase 6 Regression Test:** 10/10 passed (Zero data leakage, non-semantic shortcut audit, 91.88% test benchmark).
• **Phase 7 Production Test:** 11/11 passed (Genuine, Fake, Multi-Template, CMYK, Low-Res, Unknown Template, Corrupted Input, Large Canvas, OCR Resilience, Local Fallback, and PDF Certificate Generation).

2. Phase 1–6 Development History

• **Phase 1:** Comprehensive dataset audit of 2,222 SIDTD images; resolved 500 CMYK JPEGs, documented 10 template schemas.
• **Phase 2:** Connected-cluster group splitting (995 disjoint clusters); established frozen EfficientNet-B0 global baseline (79.69% Acc, 0.9112 AUC).
• **Phase 3:** Visual explainability with Grad-CAM and HiRes-CAM on layer features[8]; discovered micro-forgery downsampling bottleneck.
• **Phase 4:** Formulated auxiliary OCR field extraction and initial high-resolution patch verification.
• **Phase 5:** Integrated Streamlit web dashboard and automated ReportLab verification PDF certificate generation.
• **Phase 6:** Complete scientific validation: cryptographic leakage audit (0 leaks), shortcut audit (100% metadata cheat identified), trained dedicated patch model (99.38% Oracle Acc), extracted forensic features (ELA, noise, edge), calibrated multi-modal fusion, and established statistical significance (McNemar p = 1.47e-8, 5,000 bootstrap CIs).

3. Definitive Benchmark Metrics (Frozen Test Set N=320)

Configuration	Mode	Acc	Prec	Rec	F1	AUC
Phase 2 Global Baseline	Autonomous	79.69%	82.91%	77.51%	80.12%	0.9112
Dedicated Patch Model (Oracle GT)	Oracle GT	99.38%	100.0%	98.82%	99.40%	0.9999
Canonical Layout Prior Inspection	Autonomous	74.06%	70.09%	88.76%	78.33%	0.7515
Uniform Sliding Tile Inspection	Autonomous	68.75%	83.50%	50.89%	63.24%	0.8023
Phase 6 Production Fusion (Default)	**Autonomous**	**91.88%**	**95.54%**	**88.76%**	**92.02%**	**0.9780**
Phase 6 Strict Operating Point (0.5495)	Autonomous	92.19%	96.75%	88.17%	92.26%	0.9780

4. Production Pipeline Architecture & Traceability

The production pipeline strictly evaluates 5 complementary evidence streams:
1. **Standardized Loading:** Pillow RGB conversion eliminates CMYK non-visual shortcuts.
2. **Macro Visual Stream:** Frozen EfficientNet-B0 backbone produces global forgery probability (w = +3.066).
3. **Micro Local Stream:** Dedicated Patch Model (models/phase6/dedicated_patch_model.pth) inspects candidate fields at native sensor resolution (w = +0.224).
4. **Forensic Stream:** Error Level Analysis (ELA mean w = -3.984), noise residual inconsistency (w = -0.905), and Laplacian edge variance (w = +1.605) detect compression and boundary discontinuities.
5. **Semantic Stream:** EasyOCR confidence and syntactic format anomaly regex audits provide typographical priors.
6. **Zero Oracle Contamination:** Zero ground-truth bounding boxes, labels, or annotations are accessible during runtime.

5. Runtime Latency & Resource Benchmarks

Component / Subsystem	Hardware	Measured Latency	Share of Total
Model Cold Loading (All Checkpoints)	CPU	3.54 s	One-time init
Global CNN Forward Pass	CPU	67.8 ms	2.0%
Grad-CAM / HiRes-CAM Heatmaps	CPU	308.1 ms	9.0%
Dedicated Local Patch Inspection	CPU	302.4 ms	8.8%
Forensic Feature Extraction (ELA/Noise/Edge)	CPU	462.7 ms	13.4%
EasyOCR Field Extraction	CPU	1957.7 ms	56.9%
Calibrated Evidence Fusion	CPU	0.068 ms	< 0.01%
Total End-to-End Verification	**CPU**	**3.44 s**	**100.0%**

6. System Limitations & Final Submission Recommendation

• **Honest Boundary Conditions:** Primary evaluation is grounded on the 2,222 SIDTD documents. Single-character micro-inpainting remains the leading cause of false negatives (10/19 cases). Intricate guilloche patterns occasionally induce OCR character misreads (accounting for 7/7 false alarms).
• **AI-Assisted Status:** DocVerify is designed as an intelligent decision-support platform for compliance officers, not as an automated legal identity authority.
• **FINAL RECOMMENDATION:**
1. **Final Demo: READY** — Streamlit web dashboard operates flawlessly across real, fake, CMYK, and custom uploads.
2. **Final-Year Project Report: READY** — Complete Phase 1–7 empirical documentation, architecture, and code assets frozen.
3. **Research Paper Submission: READY** — Statistical significance established (p = 1.47e-8), 5,000 bootstrap CIs, zero-leakage audit, and autonomous production benchmarks rigorously validated.

Project Lead: Senior Machine Learning & Security Research Engineer

Handover Verdict: CERTIFIED PRODUCTION-READY

Institutional Project: DocVerify AI Final-Year Research Project

Final Sign-Off: October 17, 2026